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V12Bio

Virudhunagar District

Common Quarterly Examination - September 2025

Standard 12

BIOLOGY

Time: 3.00 Hrs. Marks: 70

BIO-BOTANY Marks: 35

Section - A

I. Choose the correct answer: 8×1=8

1) Consider the following statement(s).
   i) In protandrous flowers pistil matures earlier.
   ii) In protogynous flowers pistil matures earlier.
   iii) Herkogamy is noticed in unisexual flowers.
   iv) Distyly is present in primula.
   a) i and ii are correct b) ii and iv are correct
   c) ii and iii are correct d) i and iv are correct
2) Select the period for Mendel's hybridization experiments.
   a) 1856 - 1863 b) 1850 - 1870 c) 1857 - 1869 d) 1870 - 1877
3) _____ received Nobel Prize in concerning the role played by chromosomes in heredity.
   a) T. Boveri b) W.S. Sutton c) Wilhelm Roux d) Thomas Hunt Morgan
4) **Assertion (A)** : Gamma Rays are generally use to induce mutation in wheat varieties.
   Reason (R) : Because they carry lower energy to non-ionize electrons from atom.
   a) A and R are correct. b) A is correct R is wrong.
   c) A is wrong R is correct. d) A and R are wrong.
5) In which technique Ethidium Bromide is used?
   a) Southern Blotting b) Western Blotting
   c) Polymerase chain reaction d) Agrose gel Electroporosis
6) The prevention of large scale loss of biological interity
   a) Biopatent b) Bioethics c) Biosafety d) Biofuel
7) Who developed intergenic hybrid between potato and tomato called pomato?
   a) Kanta et al b) Yamada et al
   c) Horsh et al d) Merchers and Co-workers
8) Which of the given plant produces cardiac glycosides?
   a) Calotropis b) Acacia c) Nepenthes d) Utricularia

Section - B

II. Answer any FOUR questions: 4×2=8

9) Why does the zygote divides only after the division of primary endosperm cell?
10) Define dominant epistasis.
11) What is the difference between missense and nonsense mutation?
12) Write any three algae which used for the production of single cell protein.
13) What is proto cooperation?
14) What is phytoremediation? Give an example.

Section - C

III. Answer any THREE questions. Q.No. 19 is compulsory: 3×3=9

15) What is polyembroyony? How it can commercially exploited?
16) Differentiate incomplete dominance and codominance.
17) Write the significance of ploidy.
18) Write any three secondary metabolites and its plant source.
19) What is Junglone?

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Section - D

IV. Answer all the questions: 2×5=10

20) a) What is indosperm? Explain their types. (OR)
b) What are the steps involved in recombinant DNA technology?

21) a) Explain the basic concepts involved in plant tissue culture. (OR)
b) Give an account of various types of parasitism with examples.

BIO-ZOOLOGY

Marks: 35

I. Answer all the questions: 8×1=8

1) In which mode of Reproduction variations are seen
a) Asexual b) Parthenogenesis c) Sexual d) Zoospore formation

2) Colostrum is rich in
a) IgE b) IgA c) IgD d) IgM

3) A contraceptive pill prevents ovulation by
a) blocking fallopian tube
b) inhibiting release of FSH and LH
c) stimulating release of FSH and LH
d) causing immediate degeneration of released ovum

4) ABO Blood group in man is controlled by
a) Multiple alleles b) Lethal genes
c) Sex linked genes d) Y-linked genes

5) The total number of nitrogenous bases in human genome is estimated to be about
a) 3.5 million b) 35000 c) 35 million d) 3.1 billion

6) Which period was called "Age of Fishes"?
a) Permian b) Triassic c) Devonian d) Ordovician

7) Allergy involves
a) IgE b) IgG c) IgA d) IgM

8) Cyclosporin - A is an immuno suppressive drug produced from _____.
a) Aspergillus niger b) Manascus purpureas
c) Penicillium notatum d) Trichoderma polysporum

II. Answer any four of the following: 4×2=8

9) What is parthenogenesis? Give examples.
10) Expand the acronyms: (a) hCG (b) hpL
11) What are holandric genes?
12) What is called as non-sense codons in genetic code.
13) List out the major gases seems to be found in the primitive earth.
14) What is Bioremediation?

III. Answer any three of the following: 3×3=9

15) What is inhibin? State its function.
16) Mention the symptoms of Downs syndrome.
17) Differentiate - Leading stand and Lagging stand.
18) Differentiate between divergent evolution and convergent evolution with one example for each.
19) What is vaccine? What are its types?

IV. Answer the following: 2×5=10

20) Describe the structure of the human ovum with a neat labelled diagram.

(OR)

What are the strategies to be implemented in India attain total reproductive health?

21) Explain male heterogamety. **(OR)**
Explain the structure of immunoglobulin with suitable diagram.

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